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DEPT: BSC. (H.) COMPUTER SCIENCE

SUBJECT: DSE-DATA MINING (DM)

ASSIGNMENT

6. Perform density-based clustering algorithm on a downloaded dataset and evaluate the cluster quality by changing the algorithm's parameters.

Practical 6 > Q6.py > ...

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import DBSCAN
from sklearn.metrics import silhouette_score

# Load dataset
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/breast-cancer-wisconsin/wdbc.data"

columns = ["ID", "Diagnosis"] + [f"Feature_{i}" for i in range(1, 31)]

df = pd.read_csv(url, header=None, names=columns)

print("Dataset Shape:", df.shape)

# Select features
X = df.drop(["ID", "Diagnosis"], axis=1)

# Standardize data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Different DBSCAN parameters
parameters = [
    (2.5, 5),
    (2.5, 10),
    (3.0, 3),
    (3.0, 5),
    (3.5, 3)
]

results = []
best_score = -2
best_labels = None
best_params = None
```

```python
# Apply DBSCAN
for eps, min_samples in parameters:

    dbscan = DBSCAN(
        eps=eps,
        min_samples=min_samples
    )

    labels = dbscan.fit_predict(X_scaled)

    clusters = len(set(labels)) - (1 if -1 in labels else 0)
    noise = np.sum(labels == -1)

    # Calculate Silhouette Score
    mask = labels != -1

    if clusters >= 2:
        score = silhouette_score(
            X_scaled[mask],
            labels[mask]
        )
    else:
        score = None

    results.append([
        eps, min_samples, clusters, noise, score
    ])

    if score is not None and score > best_score:
        best_score = score
        best_labels = labels
        best_params = (eps, min_samples)

# Display results
result_df = pd.DataFrame(
    results,
    columns=[
        "Eps", "Min Samples", "Clusters",
        "Noise Points", "Silhouette Score"
    ]
)
```

```python
print("\nDBSCAN Results:")
print(result_df.to_string(index=False))

result_df.to_csv("DBSCAN_results.csv", index=False)

# Visualize clusters using PCA
from sklearn.decomposition import PCA

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

if best_labels is not None:

    plt.figure(figsize=(8, 5))

    plt.scatter(
        X_pca[:, 0],
        X_pca[:, 1],
        c=best_labels,
        cmap="viridis",
        s=25
    )

    plt.title("DBSCAN Clustering")
    plt.xlabel("Principal Component 1")
    plt.ylabel("Principal Component 2")
    plt.colorbar(label="Cluster")
    plt.tight_layout()

    plt.savefig("DBSCAN_clusters.png")
    plt.show()

    print("\nBest Parameters:", best_params)
    print("Best Silhouette Score:", round(best_score, 4))

print("\nClustering Completed Successfully!")

```

OUTPUT:

DBSCAN_results.csv > data

```
Eps,Min Samples,Clusters,Noise Points,Silhouette Score
2.5,5,2,224,0.1294587880212114
2.5,10,2,264,0.2940941517521553
3.0,3,2,111,0.4359611510589643
3.0,5,1,121,
3.5,3,1,69,

```

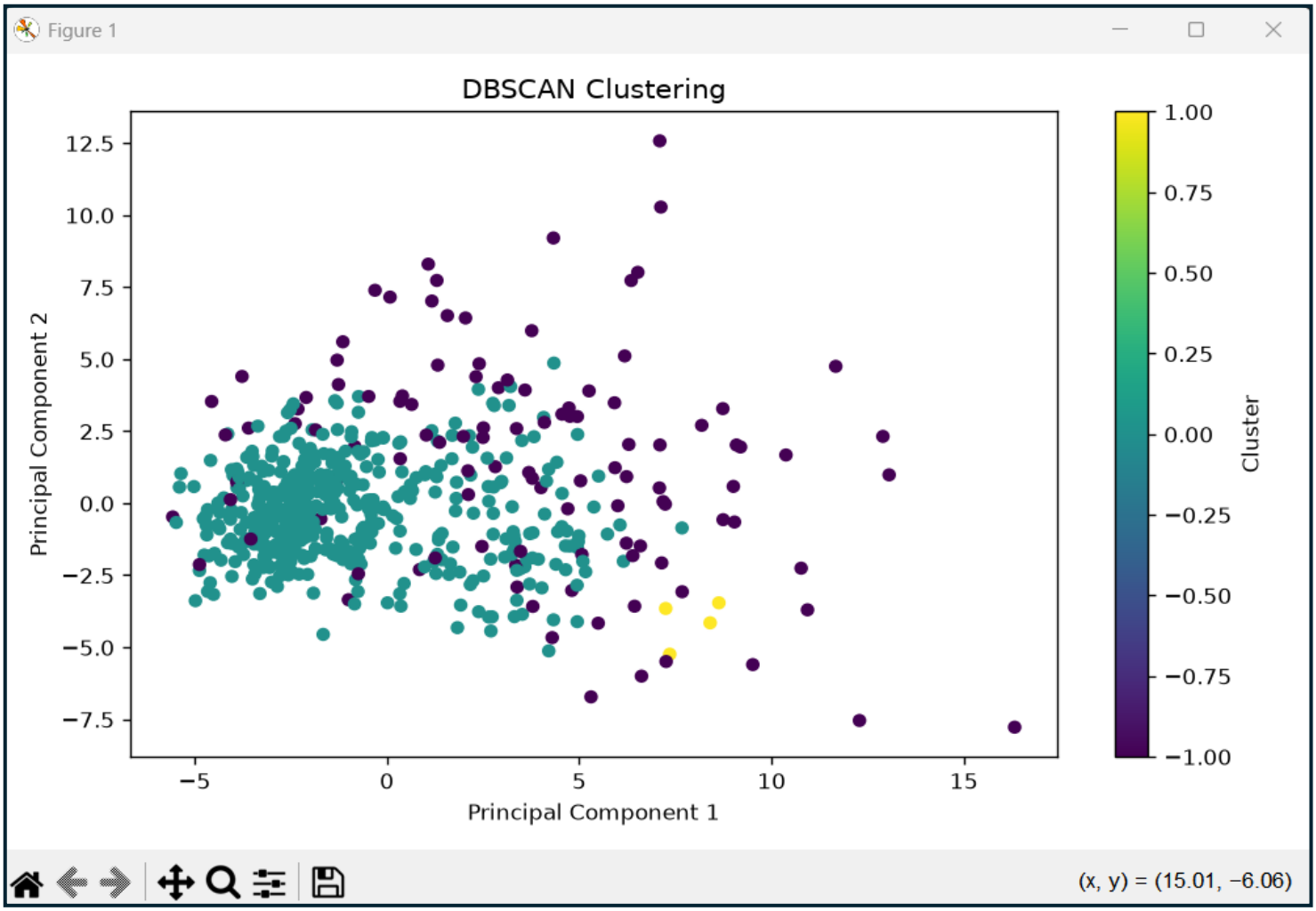
Problems Output Ports Terminal

```
PS D:\DM assignment 1> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& "d:\DM assignment 1\.venv\Scripts\Activate.ps1
(.venv) PS D:\DM assignment 1> & "d:\DM assignment 1\.venv\Scripts\python.exe" "d:/DM assignment 1/Practical 6/Q6.py"
Dataset Shape: (569, 32)

DBSCAN Results:
 Eps  Min Samples  Clusters  Noise Points  Silhouette Score
 2.5           5         2           224          0.129459
 2.5          10         2           264          0.294094
 3.0           3         2           111          0.435961
 3.0           5         1           121               NaN
 3.5           3         1            69               NaN

Best Parameters: (3.0, 3)
Best Silhouette Score: 0.436

Clustering Completed Successfully!
(.venv) PS D:\DM assignment 1>
```


Figure 1
DBSCAN Clustering
Principal Component 2
Principal Component 1
Cluster
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−5
0
5
10
15
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
(x, y) = (15.01, −6.06)

THANK YOU